CineLink — Build Plan

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Overview

CineLink is a free, browser-based, solo movie game: players build the longest chain they can by linking films and the people in them. Version 1 ships two modes, plays from a shared link on phone or desktop with no sign-up, and pulls all film data live from TMDB (The Movie Database). Head-to-head multiplayer is planned for a later phase.

	Actor Chain	Film Chain
Player is shown	An actor	A film
Player answers with	A film that actor appears in	Another film that shares a person with it
Next prompt	A different actor from the film just named (the game picks)	The film the player just named
Extra rule	Next-actor difficulty rises slowly as the chain grows	Each linking person can be used at most 4 times per run
Score	Chain length	Chain length

Goals for v1

- Playable within 30 seconds: no account, no rules page longer than three lines.
- Always current: new releases are playable as soon as TMDB lists them.
- Easy to test: one URL you can text to friends; works in mobile Safari and Chrome.
- Free to run: TMDB's free non-commercial key plus a free hosting tier.

Out of scope for v1: accounts, online multiplayer, TV shows, ads or payments.

Similar games

Film Chain is close to Cine2Nerdle's Battle mode, so this game should lead with what it does differently: solo arcade play, a shrinking timer, crew links, and the Actor Chain mode. Cine2Nerdle Battle has players alternate naming films linked by cast or crew until one runs out, and caps each person at three uses; it also uses TMDB data.

Game	How it plays	How this game differs
Cine2Nerdle Battle	Two players alternate linking films through shared cast or crew; 3 uses per person	Solo score chase first; 4-use cap; named crew roles; Actor Chain mode
Movie to Movie	Daily puzzle: link a start film to an end film in as few actor steps as possible	Endless chain, no fixed target
Link.Movie	Connect two given actors through films	Open-ended rather than A-to-B
MoviePong	Friends take turns naming actor and movie pairs, asynchronously	Solo, timed, scored

The name CineLink is distinct from all four; check the domain and social handles are free before sharing it widely.

Mode 1: Actor Chain

The player is shown an actor, names any film that actor appears in, and the game answers with another actor from that film. The run continues until the player loses three lives.

Turn by turn

1. **Start.** The game picks a well-known film at random from the start pool (see Shared systems) and shows its top-billed actor: headshot, name.
2. **Answer.** The player types a film title and picks it from the autocomplete list (poster thumbnail, title, year). Only a film picked from the list can be submitted, so typos never cost a life.
3. **Check.** The answer is correct when the shown actor has an eligible cast credit on that film and the film has not been used earlier in the run.
   - Eligible = any credited cast role, voice roles included. Archive footage and uncredited appearances do not count (see Validation).
4. **Correct.** Chain +1. The game reveals the actor's character name briefly, then picks the next actor from that film's cast:
   - never an actor already shown in this run;
   - chosen from the top N billed cast, where N grows slowly with the chain (table in Shared systems);
   - weighted towards the more popular actors in that group, using TMDB's popularity score.

5. **Wrong or timeout.** The player loses a life and the timer resets. A wrong answer keeps the same actor. A timeout swaps to a different actor from the previous film, so a blank never ends the run on its own.
6. **Dead end.** If no eligible actor is left in the top N, widen to the full credited cast. If there is still none, show a new start actor with no penalty and keep the chain count.
7. **End.** At zero lives, show the game-over screen with the score and the whole chain.

Mode 2: Film Chain

The player is shown a film and names another film that shares a person with it; the named film becomes the new starting point. The player only names the film, and the game finds the link. No person can be used as a link more than 4 times in a run.

Who counts as a link

Role	TMDB field	Rule
Actor	`cast`, using `order`	Top 15 billed: `order` 0 to 14 on **both** films (cutoff is configurable)
Director	`crew`, job `Director`	Any credit
Cinematographer	`crew`, job `Director of Photography`	Any credit
Writer	`crew`, jobs `Screenplay`, `Writer`, `Story`	Any credit; source-material credits such as `Novel` or `Characters` excluded
Composer	`crew`, job `Original Music Composer`	Any credit
Editor	`crew`, job `Editor`	Any credit

Turn by turn

1. **Start.** Show a well-known film from the start pool: poster, title, year, plus director and top 3 cast as a nudge.
2. **Answer.** The player picks a film from autocomplete, as in Actor Chain.
3. **Find links.** The game compares eligible people on the two films and drops anyone already used 4 times. The film must not have been used earlier in the run (the start film counts as used).
4. **No link left.** Wrong answer: lose a life, timer resets, same film stays.
5. **One link.** Spend it automatically.

6. **Several links.** Pause the timer and show a chooser listing each shared person, their role on each film, and uses so far (for example "Christopher Nolan, Director, 2 of 4 used"). Pre-select the person with the most uses remaining.
7. **Correct.** Chain +1; the spent person's counter goes up by 1; the named film becomes the current film.
8. **Timeout.** Lose a life; same film stays.

Counting rules

- One counter per person, whatever role they hold: a director who also wrote the film is one link, not two.
- Each correct answer spends exactly one use, even when several links matched.
- A link tray under the prompt lists every person used so far with their count (x of 4), and greys out exhausted people.

Shared systems

Both modes share one timer, one lives system and one score: 15 seconds a turn, dropping to 10 once the chain reaches 15; three lives; score = chain length.

Timer

- 15 seconds per turn until the chain reaches 15, then 10 seconds per turn for the rest of the run.
- Resets to full after every answer, right or wrong.
- Pauses while the game is checking an answer with the server, and while the Film Chain link chooser is open, so slow connections and choosing never cost time.
- Shown as a draining bar; turns amber at 5 seconds and red at 3.

Lives

- 3 per run. A wrong answer or a timeout costs one. No way to earn lives back in v1.

Score

- Score = number of correct answers in the run.
- Personal best per mode is saved on the device (browser local storage); no account needed.

Difficulty over a run

Chain length	Turn timer	Actor Chain: next actor drawn from
0 to 9	15 s	Top 3 billed

Chain length	Turn timer	Actor Chain: next actor drawn from
10 to 14	15 s	Top 5 billed
15 to 24	10 s	Top 5 billed
25 to 39	10 s	Top 8 billed
40 and over	10 s	Top 12 billed

Film Chain gets harder on its own as favourite people run out of uses, so only the timer scales there. Every number above lives in one config file so it can be tuned after testing.

Start pool ("popular starts, anything accepted")

- Prompts that start a run are drawn from roughly the 1,000 most-voted released films on TMDB, so openings are recognisable.
- Answers can be any released film on TMDB, however obscure.

Data source: TMDB API

Use the TMDB API (v3): it is free for non-commercial projects that credit TMDB, and its cast order and crew job fields map directly onto both modes' rules.

Terms that apply

- **Cost:** free for non-commercial use with attribution (TMDB developer FAQ). A project counts as commercial if its main purpose is making money for its owner; that needs a paid licence, reported at USD 149 a month for small companies (api.market). Revisit if the game ever carries ads or links to paid products.
- **Attribution:** an approved TMDB logo, less prominent than the game's own logo, plus the notice "This product uses the TMDB API but is not endorsed or certified by TMDB", placed in an About or Credits screen with a link to themoviedb.org (TMDB developer FAQ).
- **Rate limits:** the old 40-requests-per-10-seconds cap was removed in December 2019; the current ceiling is a soft limit of roughly 40 to 50 requests per second, enforced at the CDN (TMDB forum, apis.io). Back off on a 429 response.
- **Caching:** a secondary source says the terms allow caching responses for up to 6 months (apis.io). The terms page itself could not be opened here, so confirm this on themoviedb.org/api-terms-of-use when applying for the key.
- **Getting a key:** free from TMDB account settings, under API; the form asks what you are building and how you will use the data.

Endpoints the game needs

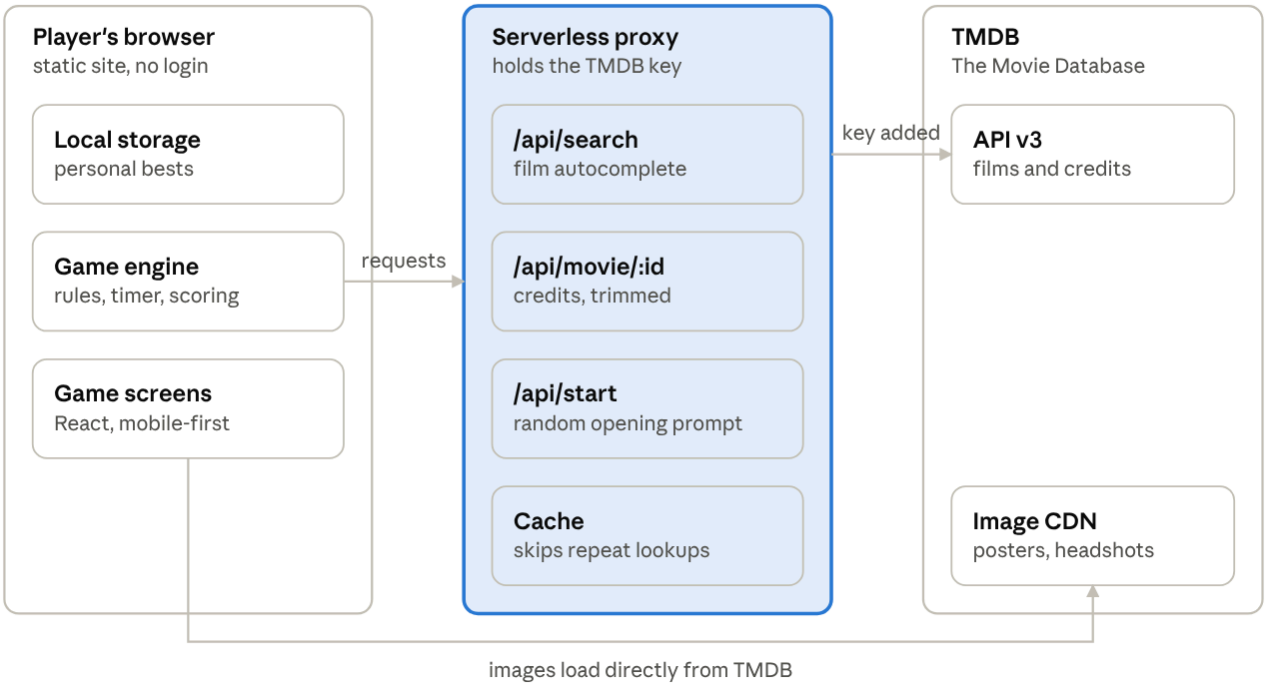
Purpose	Endpoint	Notes
Autocomplete	`GET /3/search/movie?query=…&include_adult=false`	Debounce 250 ms; show poster, title, year
Film details + credits	`GET /3/movie/{id}?append_to_response=credits`	`cast[].order` gives billing; `crew[].job` gives role
Start pool	`GET /3/discover/movie?sort_by=vote_count.desc`	Pages 1 to 50 = about 1,000 films; refresh weekly
Images	`https://image.tmdb.org/t/p/w342{poster_path}`, `/w185{profile_path}`	Loaded straight from TMDB's image CDN
Image config	`GET /3/configuration`	Base URLs and sizes; cache for a day

Crew job names (`Director`, `Director of Photography`, `Screenplay`, `Original Music Composer`, `Editor`) are as they appear in TMDB credit responses (example response). TMDB is community-edited, so some films will have missing crew; the Dispute button (Validation section) catches these.

Technical architecture

A static web app plus a thin serverless proxy, both on a free hosting tier: the browser runs the game, and the proxy holds the TMDB key, trims responses and caches them.

The TMDB key stays on the server; images load straight from TMDB



architecture · browser, proxy, TMDB

Game state lives in the browser; only lookups go through the proxy, and poster and headshot images load straight from TMDB's image CDN.

Recommended stack

Layer	Choice	Why
Front end	React + Vite + TypeScript, Tailwind CSS	Fast to build, easy for another AI to extend
Proxy	Vercel Functions or Cloudflare Workers	Free tier, deploys from the same repo, keeps the key off the client
Cache	Platform edge cache (Cloudflare KV or Vercel cache headers)	Film credits rarely change; repeat lookups skip TMDB
Hosting	Vercel or Cloudflare Pages	Every push gets a shareable preview URL for testers
Analytics	Vercel Web Analytics or Umami (free, no cookies)	See where runs end without tracking people
Feedback	Tally form linked from the game-over screen	Free, no code

Code structure

- `config.ts`: every tunable number (timers, threshold of 15, lives, top-billed cutoffs, link cap of 4, eligible crew jobs, start-pool size).
- `engine/`: pure, unit-tested functions with no network or UI code:
  - `isValidActorAnswer(actorId, filmCredits, usedFilmIds)`
  - `pickNextActor(filmCredits, usedActorIds, chainLength, config)`
  - `findLinks(creditsA, creditsB, useCounts, config)` returns every shared eligible person with their roles
  - `turnTimerSeconds(chainLength, config)`
- `api/`: proxy routes `search`, `movie/[id]`, `start?mode=actor|film`, each returning only the fields the game uses.
- `ui/`: screens and components; state in a small store such as Zustand.

Phase 2 multiplayer will need a real-time layer (for example Supabase Realtime or Cloudflare Durable Objects) and server-side answer checking; keeping the engine as pure functions lets it run on the server unchanged.

Validation and edge cases

Answers are checked by TMDB ID, never by title text, and each case below has a fixed rule so the build AI never has to guess.

Case	Rule
Films with the same title (remakes)	Autocomplete shows year and poster; the check uses the picked film's TMDB ID
Film already used this run	Rejected as wrong; the message says "Already in your chain". Applies to both modes; the start film counts as used
Typed text not picked from the list	Cannot be submitted; Enter picks the top suggestion
Archive footage or uncredited cast	Ignored: drop cast entries whose `character` contains "(archive footage)" or "(uncredited)"
Appearing as themselves (documentaries)	Accepted in v1
Unreleased films	Not accepted: `release_date` must be on or before today, so announced casts can't be used
Adult titles	Never shown: `include_adult=false` on every search

Case	Rule
TV shows	Not searchable in v1: only `/search/movie` is used
Short films and TV movies	Accepted if TMDB lists them as movies
Sequels and franchises	Allowed; the 4-use cap stops endless sequel chains in Film Chain
Missing or wrong TMDB data	A "Dispute" button on any rejected answer logs the two IDs for review. In v1 the life is still lost; disputes guide whether refunds are needed
Slow network	Timer pauses while an answer is being checked; show a spinner after 300 ms
TMDB unavailable	Show "Can't reach the film database" and pause the run rather than charging lives
Actor Chain dead end	Widen to the full cast, then fall back to a new start actor (Mode 1 rules)

Cache each film's trimmed credits on the proxy, so checking an answer and picking the next actor normally need one TMDB call or none.

Screens

Five screens cover v1, all designed phone-first in portrait with the answer box above the on-screen keyboard.

1. **Home**
   - Game name, two large buttons (Actor Chain, Film Chain), personal best under each.
   - "How to play": three lines per mode, shown in a sheet.
   - About / Credits: TMDB logo, the required notice and a link to themoviedb.org.
2. **Game**
   - Top bar: chain count, three life icons, timer bar.
   - Prompt card: headshot or poster, name or title + year. Film Chain also shows director and top 3 cast.
   - Answer box with autocomplete: poster thumbnail, title, year; up to 6 results.
   - Film Chain only: link tray of people used so far, each with "x of 4".
3. **Answer feedback** (overlay, about 1 second)

    - Correct: green flash and the link revealed, e.g. "Linked by Roger Deakins, cinematographer (1 of 4)".
    - Wrong: shake, red flash, a short reason ("Not in the cast", "No shared people left", "Already in your chain") and a Dispute button.
    - Timeout: "Time's up" and the life lost.
4. **Link chooser** (Film Chain, several links matched): list of shared people with role and uses left; timer paused.
5. **Game over**
    - Score, personal best, "New best" flag.
    - The full chain as a scrollable trail of posters and headshots, each link labelled.
    - Share: copies a short text result to the clipboard (mode, score, game link).
    - Play again, Home, and "Give feedback" (Tally form).

Style is open; a clean, cinematic look with a dark background lets posters and headshots carry the colour.

Playtesting and sharing

Testers get one link, nothing to install and no account: the hosting platform publishes every version at a public URL you can text or post in a group chat.

How to run a test round

1. Deploy to Vercel or Cloudflare Pages and share the production URL (or a preview URL for an experimental build).
2. Invite 10 to 20 people: a mix of film obsessives and casual viewers, on both iPhone and Android.
3. Ask each to play at least three runs of each mode over a week.
4. Collect feedback through the game-over form, plus anonymous play events.
5. Review disputes and events, adjust `config.ts`, ship, repeat.

Anonymous events to log (no personal data, no cookies)

- `run_start` (mode), `answer_correct` (mode, chain length, seconds taken), `answer_wrong` (mode, reason), `timeout` (mode, chain length), `dispute` (film IDs, person ID), `run_end` (mode, score).

What the round should answer

- Is the 15 to 10 second step fair, or does it end runs too abruptly?
- Typical score per mode, and whether Actor Chain runs end mostly on timeouts or wrong answers.

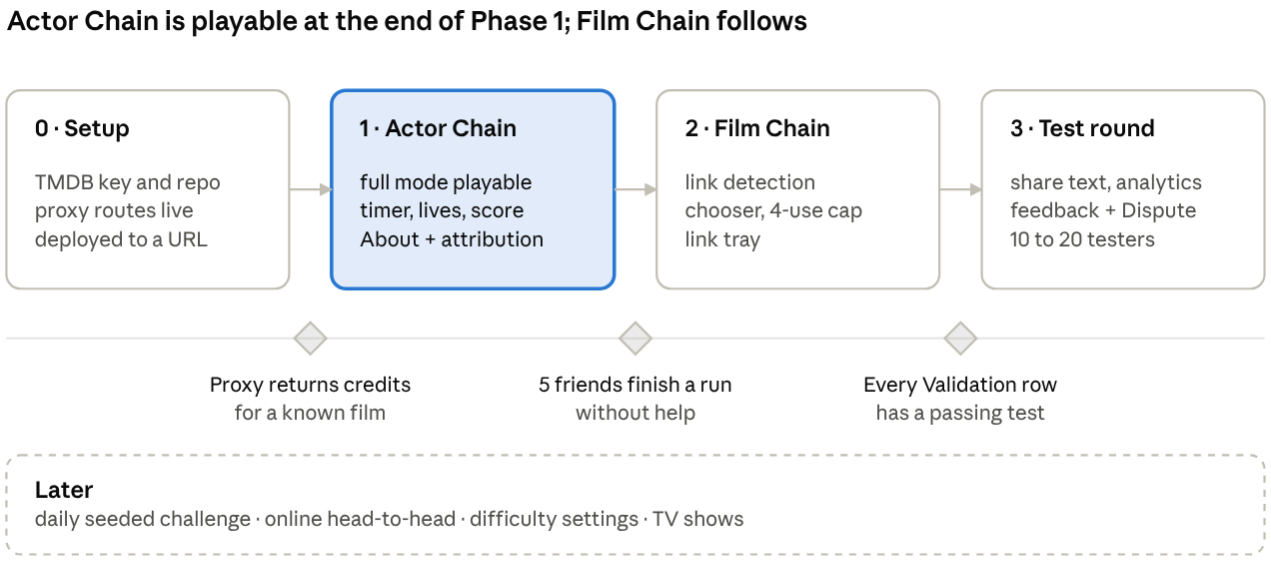
- Dispute rate: how often TMDB data lets players down.
- Do players understand the 4-use cap without reading the rules?
- Which mode do people replay more?

Feedback form questions

1. Which mode did you enjoy more, and why?
2. Did any answer get marked wrong that you're sure was right? Which?
3. Did the timer feel too short, about right, or too long?
4. Was anything confusing in the first minute?
5. Would you send this to a friend?

Roadmap

Build in four phases, each ending in something testers can play; Actor Chain ships first because it needs the simplest rules.



roadmap · 4 phases, 3 gates

Each diamond is the check that must pass before the next phase starts. No dates are set; each phase ends with a live URL.

Decisions

All rule choices are settled; this table records the ones the original brief left open.

Decision	Choice
Game name	CineLink
Actor Chain timeout	Swap to another actor from the previous film
Actor Chain: credits that count as correct	Any credited cast role
Film Chain: which actors can link	Top 15 billed (`order` 0 to 14) on both films
Writer credits	Screenplay, Writer, Story; not Novel or Characters
Person in several roles	One shared counter
"Self" credits in documentaries	Accepted
Dispute refunds a life	No; disputes are logged for review

Handoff brief for the build AI

Paste this doc into the build tool along with the prompt below.

```
You are building CineLink, a free, solo, browser-based movie linking game.
The attached plan is the spec; follow it exactly and flag anything ambiguous
instead of guessing.

Build order:
1. Phase 0: repo, React + Vite + TypeScript, serverless proxy for the TMDB v3
API (key stored as a server environment variable, never sent to the browser),
deploy to Vercel or Cloudflare Pages.
2. Phase 1: Actor Chain mode, end to end, with timer, lives, scoring, game-
over screen and the About screen with TMDB attribution.
3. Phase 2: Film Chain mode, including link detection, the link chooser and
the 4-use cap.
4. Phase 3: share text, anonymous analytics events, feedback link, Dispute
button.

Requirements:
- All tunable numbers in config.ts (timers, threshold 15, lives 3, top-billed
cutoffs, link cap 4, eligible crew jobs, start pool size).
- Game rules as pure functions in engine/ with unit tests covering every row
of the Validation table.
- Validate answers by TMDB ID only. Use include_adult=false on every search.
Movies only, released only.
- Proxy routes return trimmed JSON and cache film credits.
- Mobile-first: test at 375px wide with the on-screen keyboard open.
- Timer pauses during network checks and while the link chooser is open.

At the end of each phase, give me a live URL and a short list of what to
test.
```

Sources

- TMDB developer FAQ: cost, attribution, commercial definition
- TMDB forum on rate limits and apis.io TMDB limits: rate limits, caching
- api.market movie API comparison: commercial licence price
- TMDB credits response example: cast order and crew job fields
- Jinan's Puzzle Nook and Polygon via blog repost: Cine2Nerdle Battle rules
- Movie to Movie, Link.Movie, MoviePong: similar games